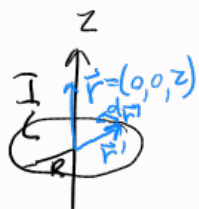


4.3) Mag. polje krožne zanke vzdolž osi, pokaži da je $\int_{-\infty}^{\infty} \vec{H}_z(z) dz = I$



• Biot-Savart: $\vec{H}(\vec{r}) = \frac{I}{4\pi} \int \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$

• Ampere: $\oint \vec{H} \cdot d\vec{l} = I$

$$\vec{r} - \vec{r}' = (-R \cos \varphi, -R \sin \varphi, z)$$

$$d\vec{r}' = \frac{\partial \vec{r}'}{\partial \varphi} d\varphi = (-R \sin \varphi, R \cos \varphi, 0) d\varphi$$

$$|\vec{r} - \vec{r}'| = \sqrt{R^2 + z^2}$$

$$\rightarrow \vec{H}(\vec{r}) = \frac{I}{4\pi} \int \frac{R^2 (-\sin \varphi, \cos \varphi, 0) \times (-\cos \varphi, -\sin \varphi, z/R)}{(R^2 + z^2)^{3/2}} d\varphi =$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin \varphi & \cos \varphi & 0 \\ -\cos \varphi & -\sin \varphi & z/R \end{vmatrix} = \begin{bmatrix} z/R \cos \varphi \\ z/R \sin \varphi \\ 1 \end{bmatrix}$$

$$= \frac{I}{4\pi} \frac{R^2}{(R^2 + z^2)^{3/2}} \int_0^{2\pi} (z/R \cos \varphi, z/R \sin \varphi, 1) d\varphi = \frac{I}{4\pi} \frac{R^2}{(R^2 + z^2)^{3/2}} (0, 0, 2\pi) = \frac{I R^2}{2(R^2 + z^2)^{3/2}} (0, 0, 1)$$

$$H_z(z) = \frac{I R^2}{2(R^2 + z^2)^{3/2}}$$

$$z = R \tan u \\ dz = \frac{R du}{\cos^2 u}$$

$$\int_{-\infty}^{\infty} H_z dz = \frac{I R^2}{2} \int_{-\infty}^{\infty} \frac{1}{(R^2 + z^2)^{3/2}} dz = \frac{I R^3}{2} \int_{-\pi/2}^{\pi/2} \frac{1}{(R^2 + R^2 \tan^2 u)^{3/2} \cos^2 u} du =$$

$$= \frac{I}{2} \int_{-\pi/2}^{\pi/2} \frac{1}{(1 + \tan^2 u)^{3/2}} du = \frac{I}{2} \sin u \Big|_{-\pi/2}^{\pi/2} = I$$

Bronštejn: $x = ax^2 + bx + c, \Delta = 4ac - b^2$

$$\int \frac{dx}{x^{3/2}} = \frac{2(2ax + b)}{\Delta \sqrt{x}}$$

4.1) Magn. polje polneskrajne žice



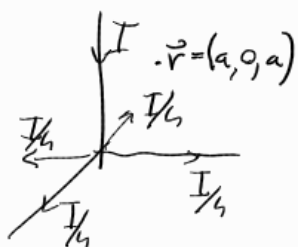
$$\vec{r}' = t \hat{e}_x$$

$$d\vec{r}' = dt \hat{e}_x$$

$$\vec{H}(\vec{r}) = \frac{I}{4\pi} \int \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} = \frac{I}{4\pi} \int \frac{\hat{e}_x \times (\vec{r} - t \hat{e}_x)}{(\sqrt{r^2 - 2t \vec{r} \cdot \hat{e}_x + t^2})^3} dt = \left(\frac{1}{|\vec{e}_x \times \vec{r}|^2} \left(1 + \frac{\vec{r} \cdot \hat{e}_x}{r} \right) \right)$$

$$= \frac{I}{4\pi} (\hat{e}_x \times \vec{r}) \int_0^{\infty} \frac{dt}{(r^2 - 2t \vec{r} \cdot \hat{e}_x + t^2)^{3/2}} = \frac{I}{4\pi} (\hat{e}_x \times \vec{r}) \frac{2(2ax + b)}{\Delta \sqrt{x}} \Big|_0^{\infty} = \frac{I}{4\pi} (\hat{e}_x \times \vec{r}) \left[\frac{1}{(r^2 - \vec{r} \cdot \hat{e}_x)} + \frac{\vec{r} \cdot \hat{e}_x}{(r^2 - \vec{r} \cdot \hat{e}_x) |r|} \right]$$

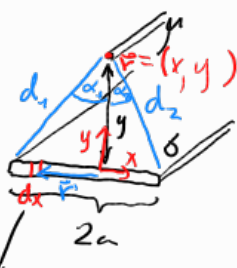
2. kolokvij 2011/2012



$$\begin{aligned}
 H(\vec{r}) &= -H(\hat{e}_z) + \frac{1}{4} (H(\hat{x}) + H(-\hat{x}) + H(\hat{y}) + H(-\hat{y})) = \\
 &= \frac{I}{16\pi} \left[-\frac{\hat{e}_z \times \vec{r}}{|\hat{e}_z \times \vec{r}|^2} \left(1 + \frac{\vec{r} \cdot \hat{e}_z}{r}\right) + \frac{\hat{x} \times \vec{r}}{|\hat{x} \times \vec{r}|^2} \left(1 + \frac{\vec{r} \cdot \hat{x}}{r}\right) + \right. \\
 &\quad \left. - \frac{\hat{x} \times \vec{r}}{|\hat{x} \times \vec{r}|^2} \left(1 - \frac{\vec{r} \cdot \hat{x}}{r}\right) + \frac{\hat{y} \times \vec{r}}{|\hat{y} \times \vec{r}|^2} \left(1 + \frac{\vec{r} \cdot \hat{y}}{r}\right) - \frac{\hat{y} \times \vec{r}}{|\hat{y} \times \vec{r}|^2} \left(1 - \frac{\vec{r} \cdot \hat{y}}{r}\right) \right] = \\
 &= \frac{I}{16\pi} \left[-\frac{4}{a} \left(1 + \frac{a}{a\sqrt{2}}\right) - \frac{1}{a} \left(1 + \frac{a}{a\sqrt{2}}\right) + \frac{1}{a} \left(1 - \frac{a}{a\sqrt{2}}\right) \right] = \\
 &= \frac{I}{16\pi a} \left[-4 - \frac{4}{\sqrt{2}} - 1 - \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} \right] = \\
 &= \frac{I}{16\pi a} [-4 - 3\sqrt{2}]
 \end{aligned}$$

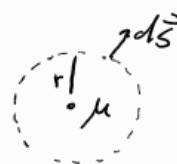
2. kolokvij 2010/2011

Sila med nabitim trakom in nabito žico



$$\vec{F}/L = ?$$

• El. polje nabite žice: Gauss: $\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$



$$\rightarrow 2\pi r L E(r) = \frac{\mu L}{\epsilon_0} \rightarrow E(\vec{r}) = \frac{\mu}{2\pi\epsilon_0} \frac{\vec{r}}{r^2}$$

$$\begin{aligned}
 d\vec{E} &= \frac{dq}{2\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} = \frac{\sigma dx}{2\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} \\
 \vec{r} &= (x, y) \\
 \vec{r}' &= (x', 0)
 \end{aligned}$$

$$\vec{E} = \frac{\sigma}{2\pi\epsilon_0} \int_{-a}^a \frac{(x-x', y)}{(x-x')^2 + y^2} dx'$$

$$\begin{aligned}
 (x-x')^2 + y^2 &= u \\
 -2(x-x')dx' &= du \\
 \frac{x-x'}{y} &= v \\
 \frac{-dx'}{y} &= dv
 \end{aligned}$$

$$E_x = \frac{\sigma}{2\pi\epsilon_0} \int_{u_1}^{u_2} \left(-\frac{1}{2}\right) \frac{du}{u} = \frac{\sigma}{4\pi\epsilon_0} \ln \frac{u_1}{u_2} = \frac{\sigma}{4\pi\epsilon_0} \ln \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} = \frac{\sigma}{2\pi\epsilon_0} \ln \frac{d_1}{d_2}$$

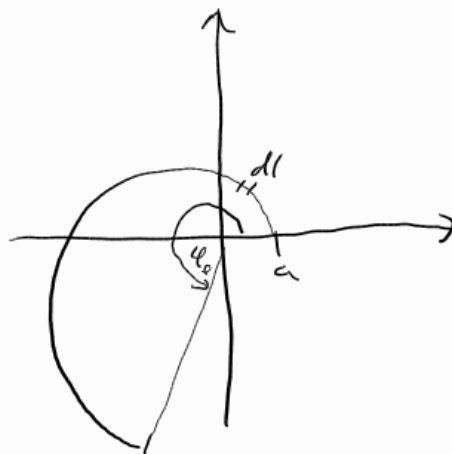
$$\begin{aligned}
 E_y &= \frac{\sigma}{2\pi\epsilon_0} \int_{-a}^a \frac{y dx'}{(x-x')^2 + y^2} = -\frac{\sigma}{2\pi\epsilon_0} \int_{v_1}^{v_2} \frac{dv}{1+v^2} = -\frac{\sigma}{2\pi\epsilon_0} \arctan v \Big|_{v_1}^{v_2} = \frac{\sigma}{2\pi\epsilon_0} [\arctan(\frac{x+a}{y}) - \arctan(\frac{x-a}{y})] = \\
 &= \frac{\sigma}{2\pi\epsilon_0} [\alpha_1 + \alpha_2]
 \end{aligned}$$

2. kol. 2015

habita spirala oblike $r = ae^{k\varphi}$, dolžine l

$$V(\vec{r}=0) = ?$$

$$\vec{E}(\vec{r}=0) = ?$$



$$\vec{r} = ae^{k\varphi} \hat{e}_r \rightarrow |\vec{r}| = ae^{k\varphi}$$

$$d\vec{r} = ake^{k\varphi} \hat{e}_r d\varphi + ae^{k\varphi} \hat{e}_\varphi d\varphi = ae^{k\varphi} (k\hat{e}_r + \hat{e}_\varphi) d\varphi$$

$$dl = |d\vec{r}| = ae^{k\varphi} \sqrt{k^2 + 1} d\varphi$$

$$dV = \frac{\mu d\vec{r}}{4\pi\epsilon_0 |\vec{r}|^3} \rightarrow V = \frac{\mu}{4\pi\epsilon_0} \int_0^{\varphi_0} \frac{ae^{k\varphi} \sqrt{k^2 + 1} d\varphi}{ae^{k\varphi}} = \frac{\mu \sqrt{k^2 + 1} \varphi_0}{4\pi\epsilon_0} \quad \checkmark$$

$$l = \int_0^{\varphi_0} |d\vec{r}| = \int_0^{\varphi_0} ae^{k\varphi} \sqrt{k^2 + 1} d\varphi = \frac{a}{k} \sqrt{k^2 + 1} (e^{k\varphi_0} - 1)$$

$$\vec{E} = \frac{e}{4\pi\epsilon_0 r^2} \hat{e}_r \quad (\text{točkast naboj})$$

$$\rightarrow \vec{E} = -\frac{\mu}{4\pi\epsilon_0} \int \frac{|d\vec{r}| \vec{r}}{|\vec{r}|^3} = -\frac{\mu}{4\pi\epsilon_0} \int_0^{\varphi_0} \frac{ae^{k\varphi} \sqrt{k^2 + 1} d\varphi}{(ae^{k\varphi})^3} ae^{k\varphi} \hat{e}_r =$$

$$= -\frac{\mu \sqrt{k^2 + 1}}{4\pi\epsilon_0 a} \int_0^{\varphi_0} \frac{d\varphi}{e^{k\varphi}} \quad \left(\begin{matrix} \text{ } \\ \text{ } \end{matrix} \right) \quad \vec{E} = E_x + iE_y$$

$$\Rightarrow \vec{E} = -\frac{\mu \sqrt{k^2 + 1}}{4\pi\epsilon_0 a} \int_0^{\varphi_0} e^{-k\varphi} e^{i\varphi} d\varphi = -\frac{\mu \sqrt{k^2 + 1}}{4\pi\epsilon_0 a} \frac{1}{i-k} \left[e^{(i-k)\varphi_0} - 1 \right]$$

$$\frac{1(-i-k)}{(i-k)(-i-k)} = -\frac{i+k}{1+k^2}$$

$$= + \frac{\mu_0}{4\pi\epsilon_0 a \sqrt{1+k^2}} \left[(i+k) \left(e^{-k\varphi_0} (\cos\varphi_0 + i \sin\varphi_0) - 1 \right) \right] =$$

$$= \frac{\mu_0}{4\pi\epsilon_0 a \sqrt{1+k^2}} \left(-k + e^{-k\varphi_0} (k \cos\varphi_0 + \sin\varphi_0), e^{-k\varphi_0} (\cos\varphi_0 + k \sin\varphi_0) - 1 \right)$$

POLARNA PARAMET. ELIPSE: $|\vec{r}| = \frac{p}{1 + \epsilon \cos\varphi}$; $p = \frac{b^2}{a}$ in $\epsilon = 1 - \frac{b^2}{a^2}$
 (izhališče i gorišču) ↓ ↓
 griščni parameter ekscentričnost